# DG Interns Hub – Week 1

## Lab Setup, Linux and Networking Foundations

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Program: DG Interns Hub – Week 1

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## 1. Folder Structure

```
DG Interns Hub
 |-- Week 1
        |-- Notes      (8 theory answers, one file per question)
        |-- Writeups     (TryHackMe room writeups)
        |-- Findings
        |   |-- Lab Setup    (VM list, IPs, pings, snapshots)
        |   |-- Linux Practice  (users, groups, permissions, commands, cron)
        |   |-- Network Map   (network commands + draw.io diagram)
        |   |-- Over the Wire Bandit
        |-- Cheat Sheet   (Linux Command - What It Does - When to Use It)
        |-- PDF Report
        |-- Presentation
        |-- README.md
```

## 2. Lab Setup

| VM | Hypervisor | Network | IP | Gateway |
|---|---|---|---|---|
| Parrot Security 7.2 (used in place of Kali, approved by instructor) | VMware Workstation | NAT | 192.168.32.128/24 (ens33) | 192.168.32.2 |
| Ubuntu | VirtualBox | NAT | 10.0.2.15/24 (enp0s3) | 10.0.2.2 |

- A clean snapshot named Clean was taken for each VM.
- Screenshots: VM lists, IP addresses, internet pings and snapshots are in `Findings/Lab Setup`.

## 3. Linux Practice

Performed on Parrot (all commands and outputs are in `Findings/Linux Practice`); users and groups were also created on Ubuntu.

- 3 users: alice, bob, carol
- 2 groups: `devteam` (alice, bob) and `analysts` (carol)
- Shared folder: `/srv/shared`, owner `root`, group `devteam`, mode `2770` (`drwxrws---`). alice and bob can create fles; carol gets Permission denied.
- Commands used: `find /etc -name "*.conf"`, `grep "root" /etc/passwd`, `ps aux`, `sudo ss -tulpn`
- Cron job: `* * * * * echo "hello $(date)" >> /tmp/cron.log` (output verifed with `cat /tmp/cron.log`)

## 4. Network Map

Recorded with `ip a`, `ip route`, `cat /etc/resolv.conf`, `ping`, `traceroute`, `nslookup` / `dig` on both VMs.

| VM | IP | Gateway | DNS |
|---|---|---|---|
| Parrot (VMware) | 192.168.32.128 | 192.168.32.2 | see `parrot_02_ip_route_resolv` screenshot |
| Ubuntu (VirtualBox) | 10.0.2.15 | 10.0.2.2 | 192.168.100.1 (local stub 127.0.0.53) |

The diagram (draw.io) is in `Findings/Network Map` as PNG and `.drawio`. The two VMs run on different hypervisors, so each reaches the internet through its own NAT.

## 5. Cheat Sheet

A table "Linux Command - What It Does - When to Use It" with 42 entries is in the `Cheat Sheet` folder.

## 6. Notes (Theory)

Eight separate fles covering cybersecurity and the CIA triad, SOC vs VAPT vs incident responder, OSI vs TCP/IP, IP/MAC/DNS/DHCP and ports, Linux fle system and permissions, threat/vulnerability/risk/exploit, VM networking modes, and legal and ethical basics.

## 7. TryHackMe Writeups

| # | Room | Status |
|---|---|---|
| 1 | Search Skills | Completed (100%) |
| 2 | Learning Cyber Security | Completed (100%) |
| 3 | Linux Fundamentals Part 1 | Completed (100%) |
| 4 | What is Networking? | Completed (100%) |
| 5 | Introductory Networking | Completed (100%) |
| 6 | Networking Concepts | Completed (100%) |
| 7 | Pyramid of Pain | Completed (100%) |
| 8 | Intro to LAN | Completed (100%) |
| Bonus | Defensive Security Intro | Completed (100%) |

## 8. OverTheWire Bandit (levels 0-10)

Writeups and screenshots go in `Writeups` and `Findings/Over the Wire Bandit`. Level passwords are intentionally not included anywhere.

## 9. Key Learnings

- How to build a small home lab with two VMs and read their network settings.
- Managing users, groups and permissions in Linux, including the setgid bit on shared folders.
- How IP, gateway and DNS ft together, and how ping, traceroute and dig help to check them.
- The OSI and TCP/IP models and how TCP and UDP differ.
- How defenders think (Pyramid of Pain, SOC work).